21-242 Recitation 12

Linear Transformations — Q1. Definition

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

State the definition of a linear transformation.

Course-level solution

Definition. A map $T : V \to W$ is a *linear transformation* if

$$T(a\vec{v} + b\vec{w}) = a\,T(\vec{v}) + b\,T(\vec{w}) \qquad \text{for all } \vec{v}, \vec{w} \in V \text{ and all } a, b \in \mathbb{F}.$$

Equivalently: $T(\vec{v} + \vec{w}) = T(\vec{v}) + T(\vec{w})$ and $T(a\vec{v}) = a\,T(\vec{v})$.